



Networking Innovations for Gigascale AI Systems

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Hot Interconnect 2026



NVIDIA Vera Rubin Extreme Co-Design

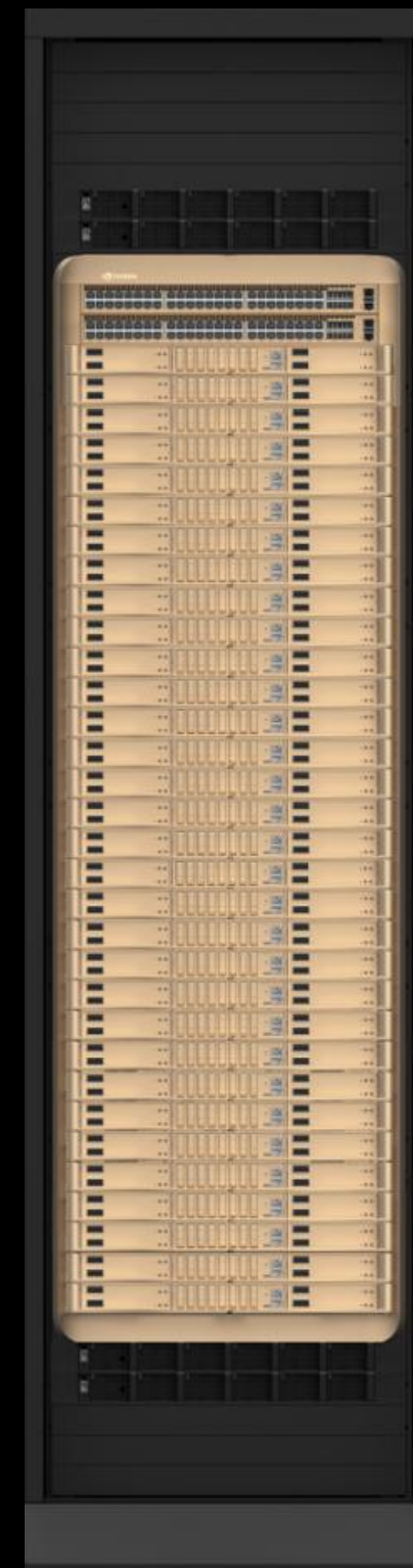
Vera Rubin
NVL72



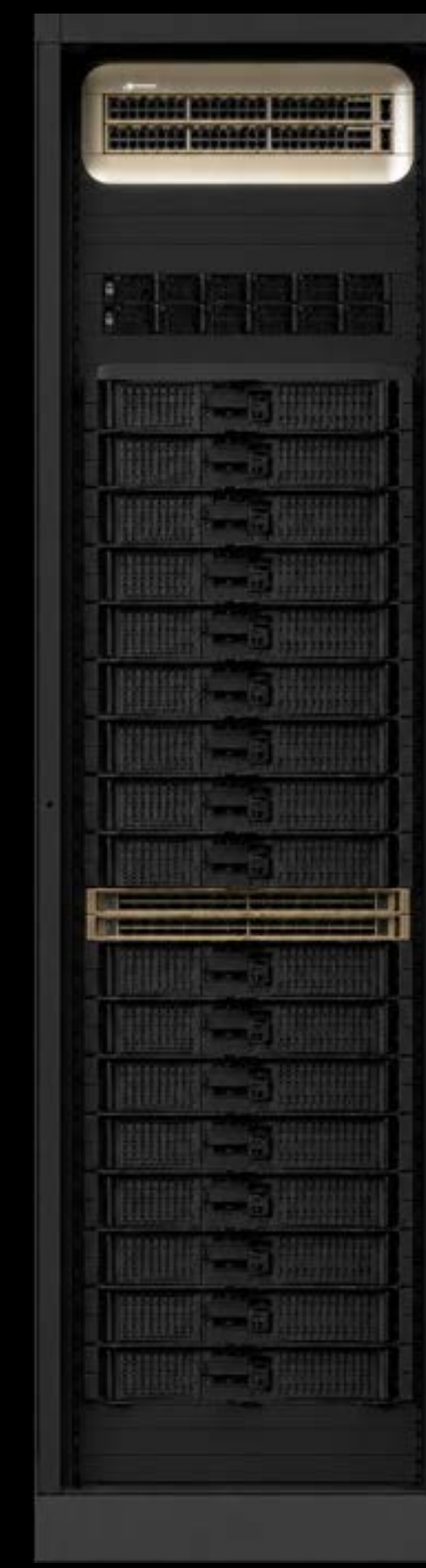
Groq 3 LPX



Vera CPU
Rack



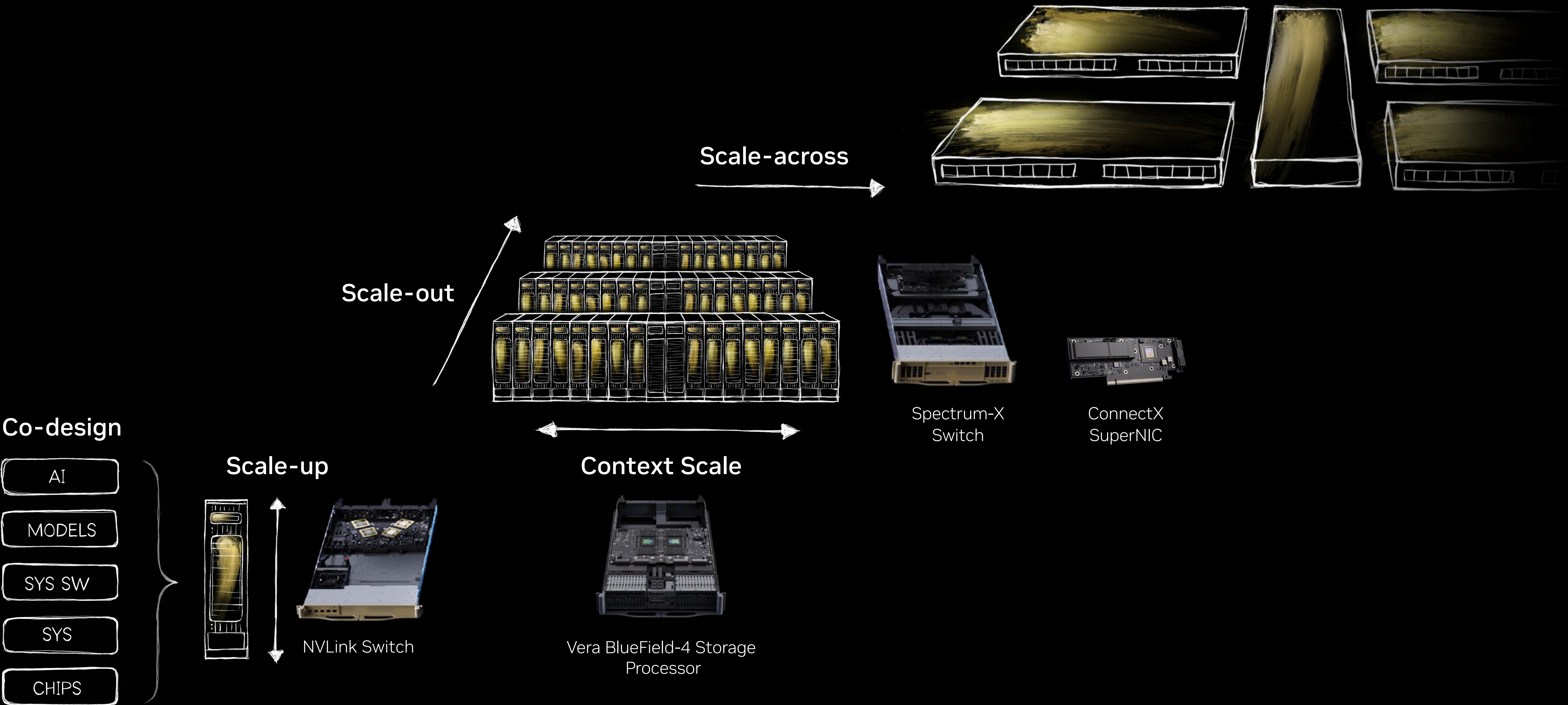
Vera BlueField-4
STX Storage



Spectrum-6 SPX
Networking



The Networking Infrastructures of NVIDIA AI Factory



NVIDIA NVLink Scale-up Networking Fabric for AI Factories

Purpose-built scale-up fabric delivers leading performance and intelligent resiliency

NVL72 Rack

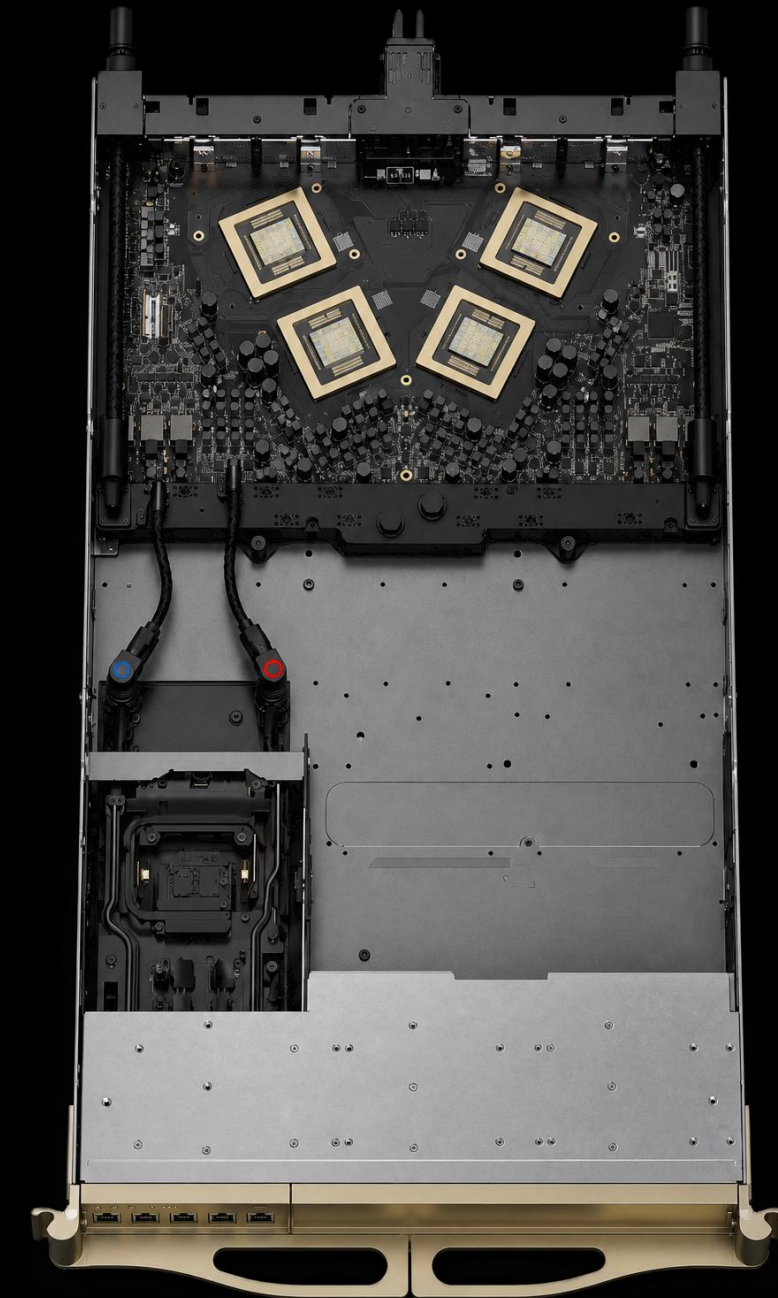
72 GPU scale-up domain



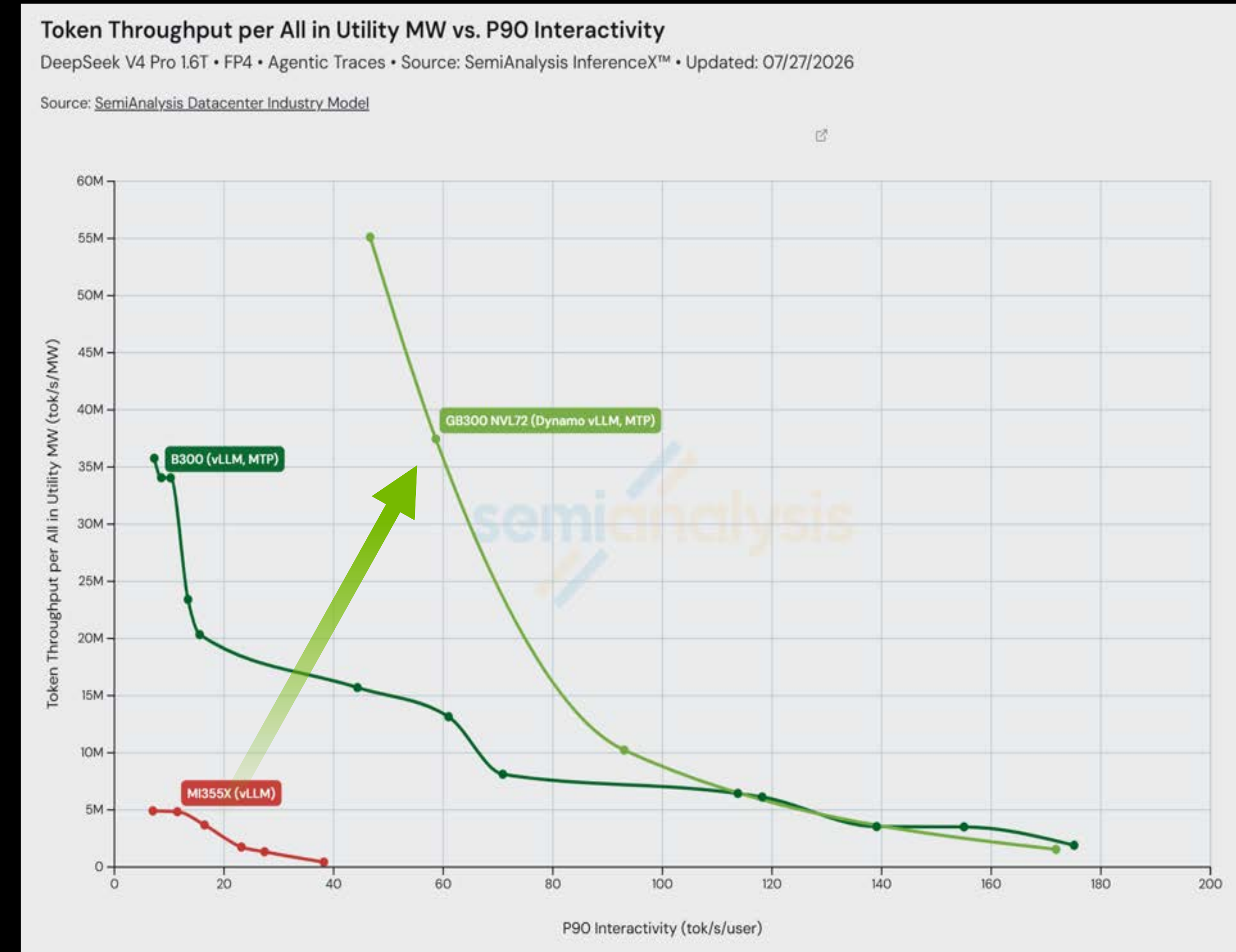
NVLink Spine



NVLink Switch Tray

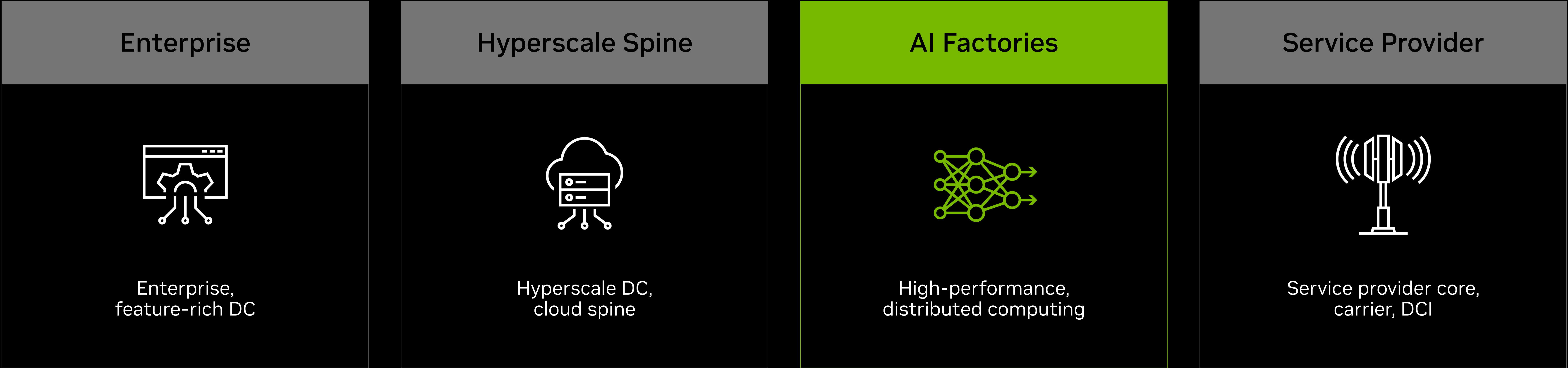


NVLink NVL72 Enables Leading tokens/MW



The Different Ethernet Architectures

AI Factories require purpose-built infrastructure



Spectrum-X Purpose-Built Scale-Out Ethernet for AI Factories

Scaling AI factories of hundreds of thousands of GPUs

The only purpose-built scale-out Ethernet for AI workloads

Adaptive routing, congestion control, jitter-free communications

End-to-end with 102.4T switch systems and 1.6T SuperNICs

Large ecosystem adoption and open operating systems

1.6x

Higher
RDMA bandwidth

2.2x

Multi-tenancy
bandwidth

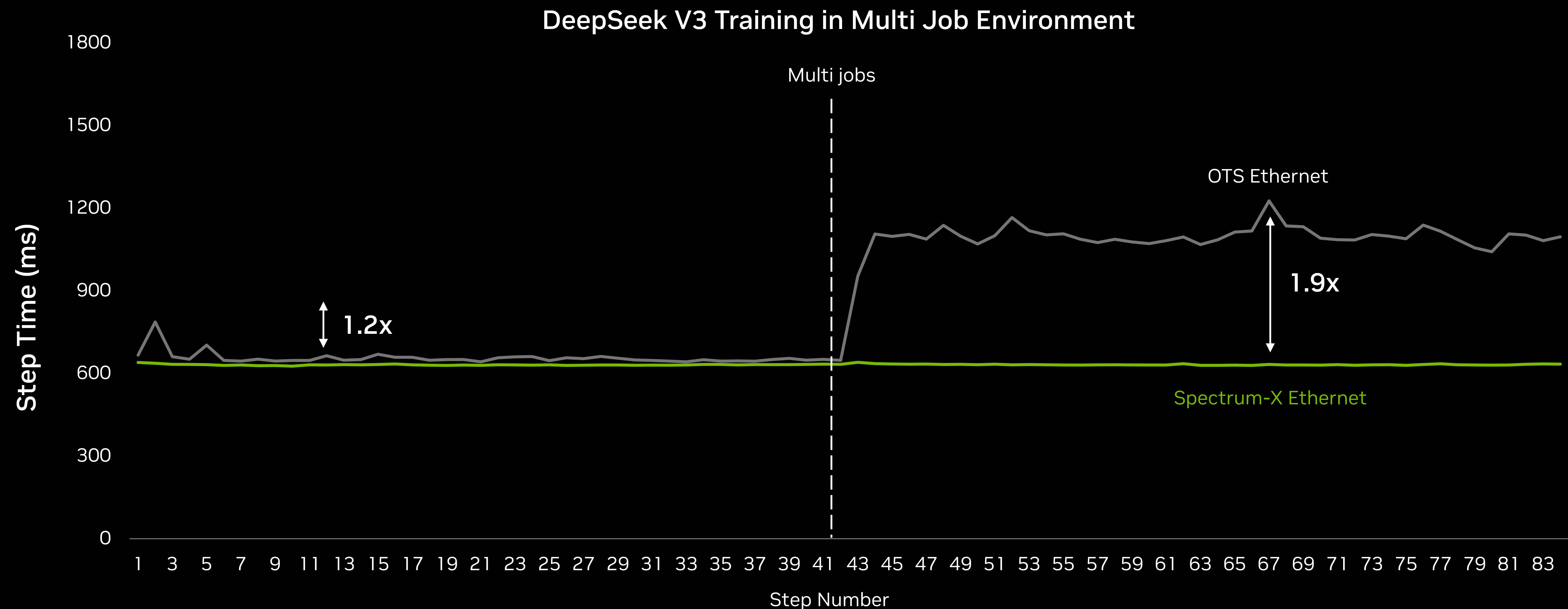
1.3x

Lower
jitter



1.9x Higher Training Performance in Multi-Tenant AI Factory

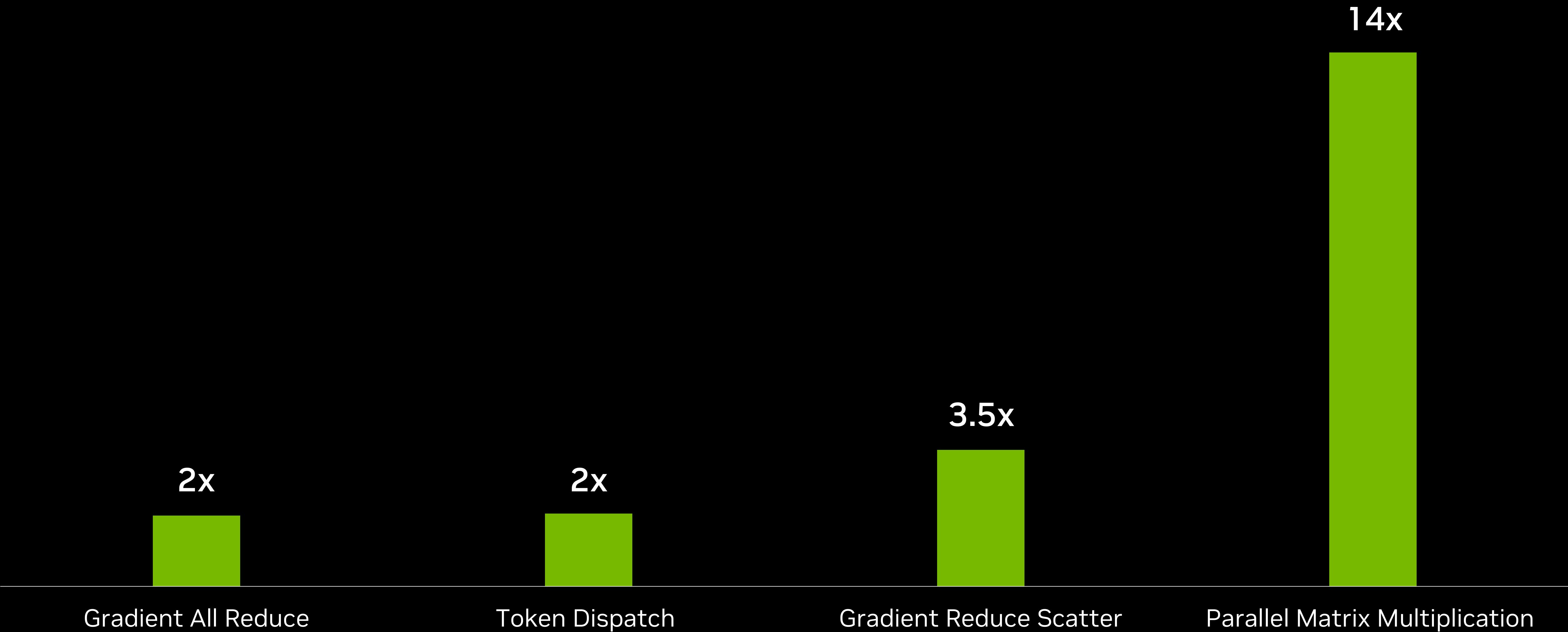
Spectrum-X Ethernet low-jitter communications, multi-tenant noise isolation
Higher performance with deterministic results



Up to 14x Higher NCCL Performance

The main NCCL operations for Nemotron Ultra pre-training

Spectrum-X Ethernet Extreme Co-Design
Performance Optimizations



Scale-Out AI Depends on Optics

Equal to 10% of compute power



Traditional Cloud Data Center



AI Factory

Spectrum-X Ethernet Photonics Increases AI Capacity and Productivity

CPO chip with micro ring modulator technology in production
3D-stacked silicon photonics engine with TSMC Coupe process
High-power, high-efficiency lasers
Detachable fiber connectors

4x

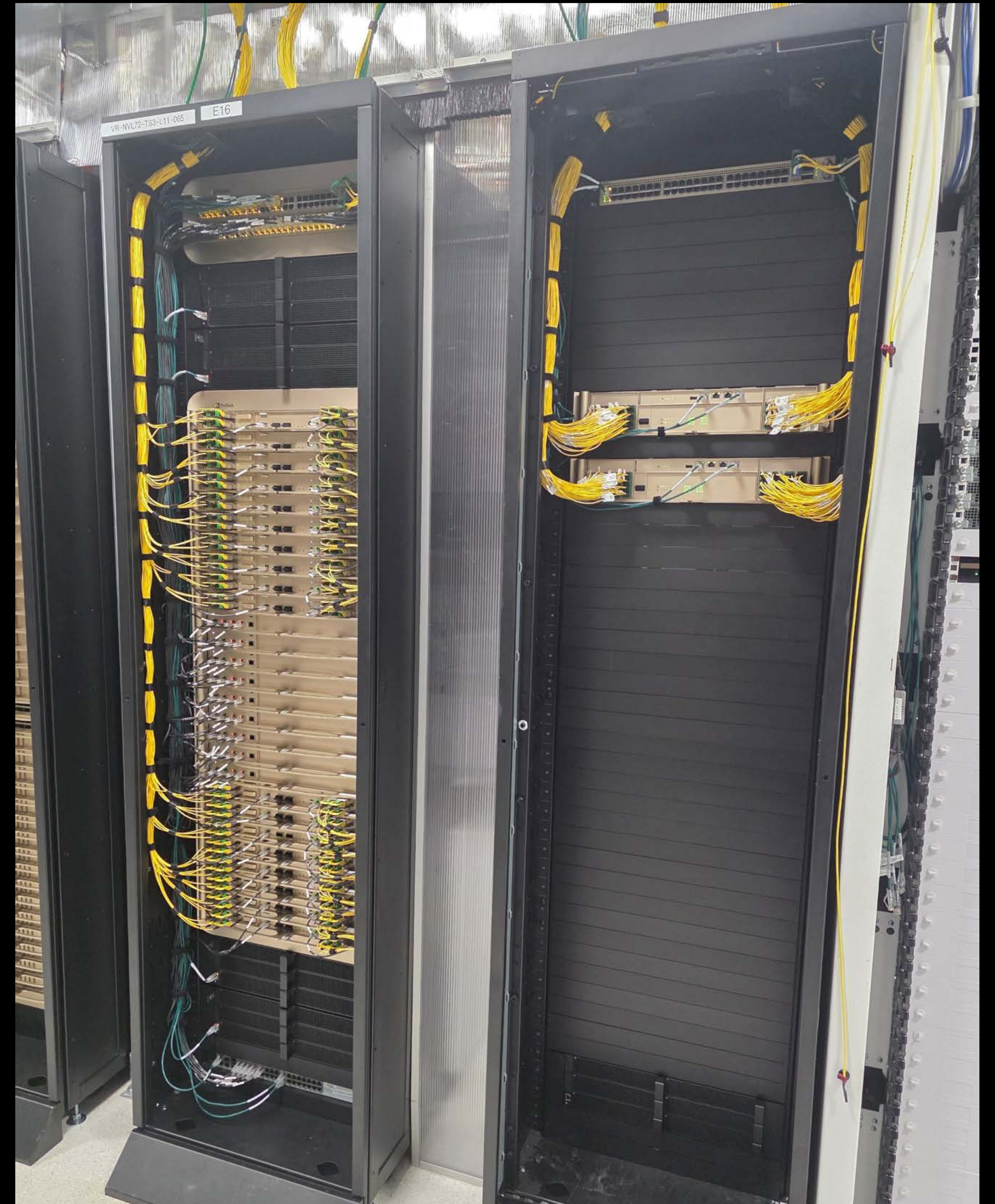
Less lasers

5x

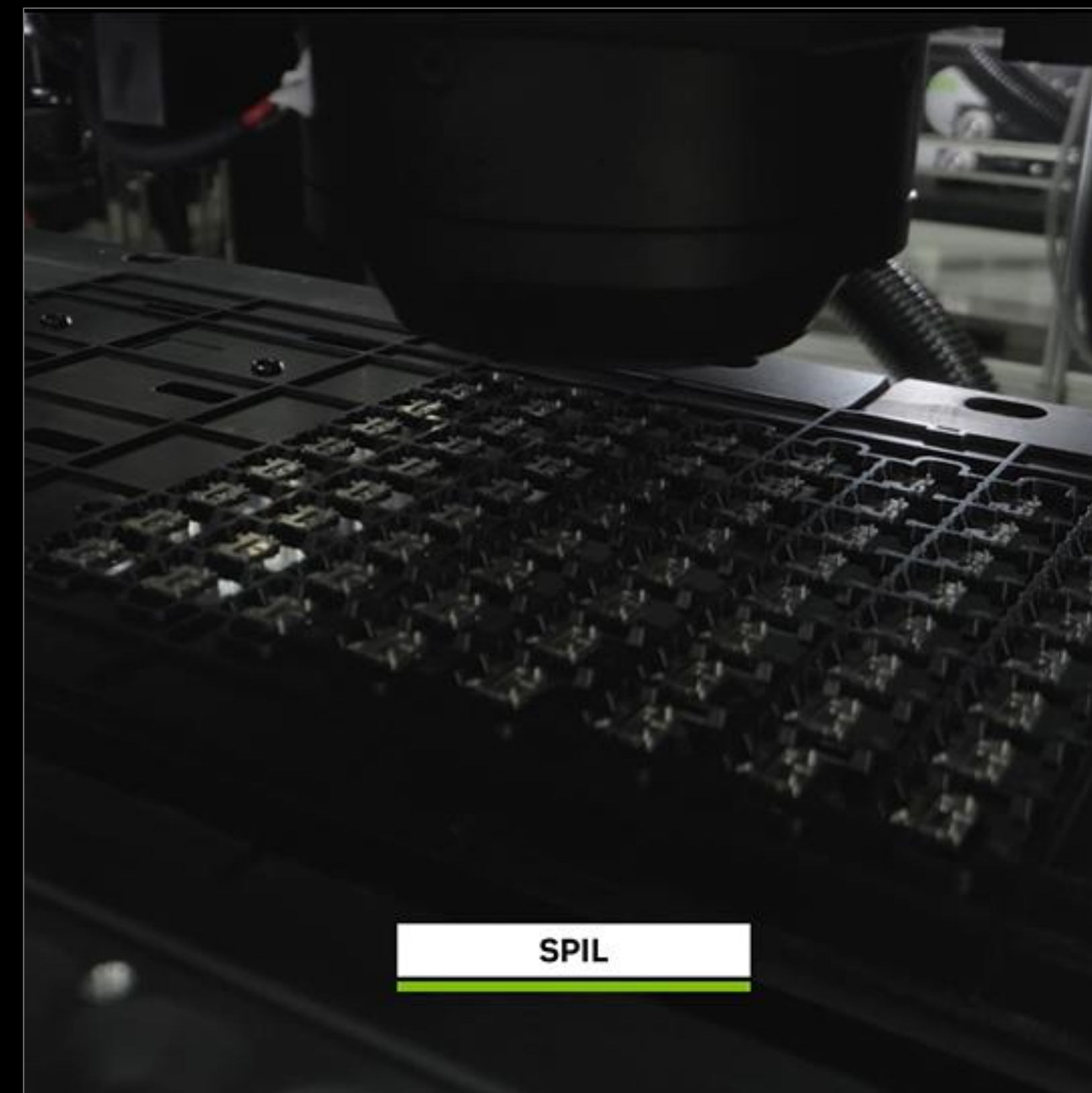
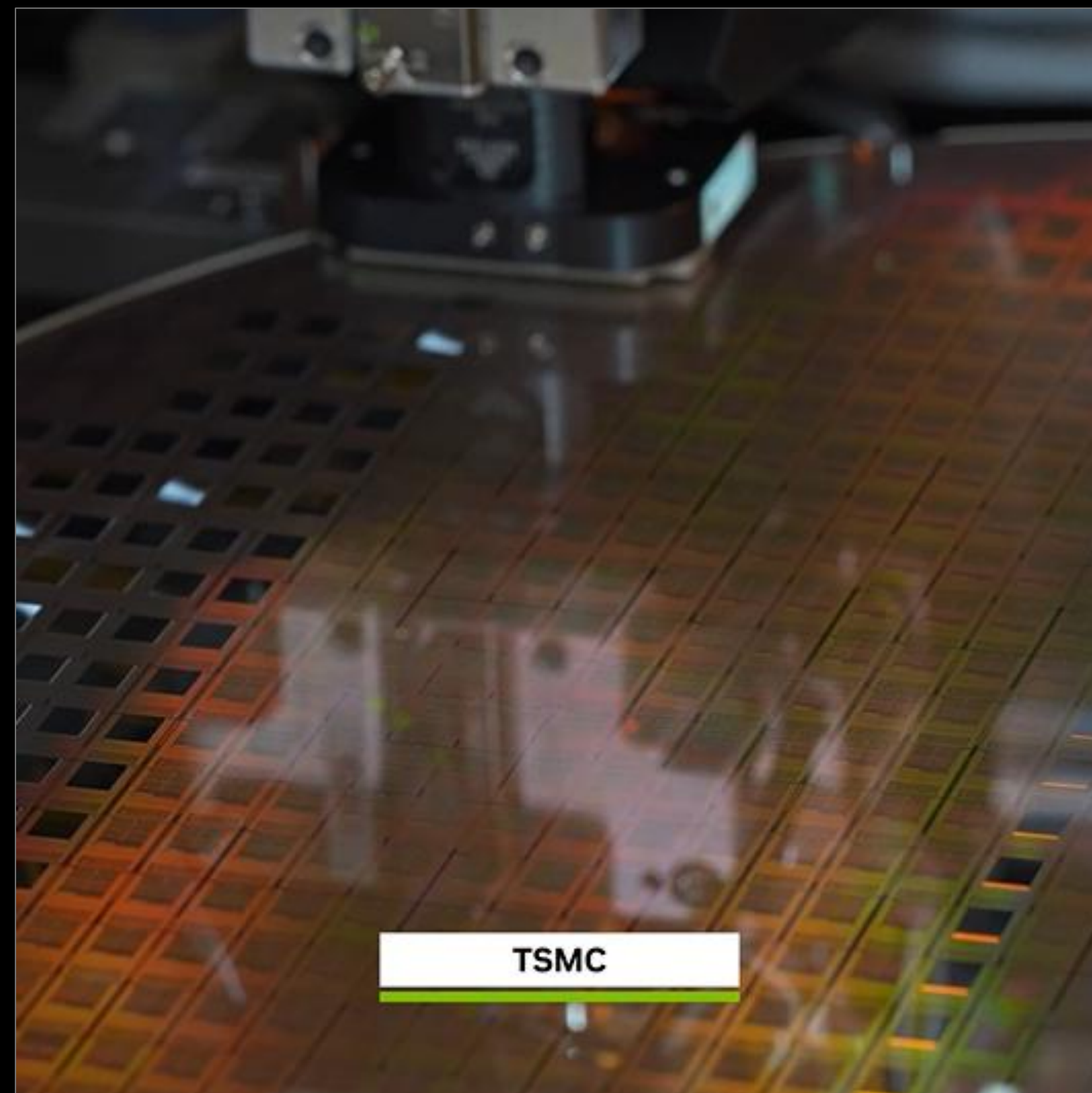
Lower power

10x

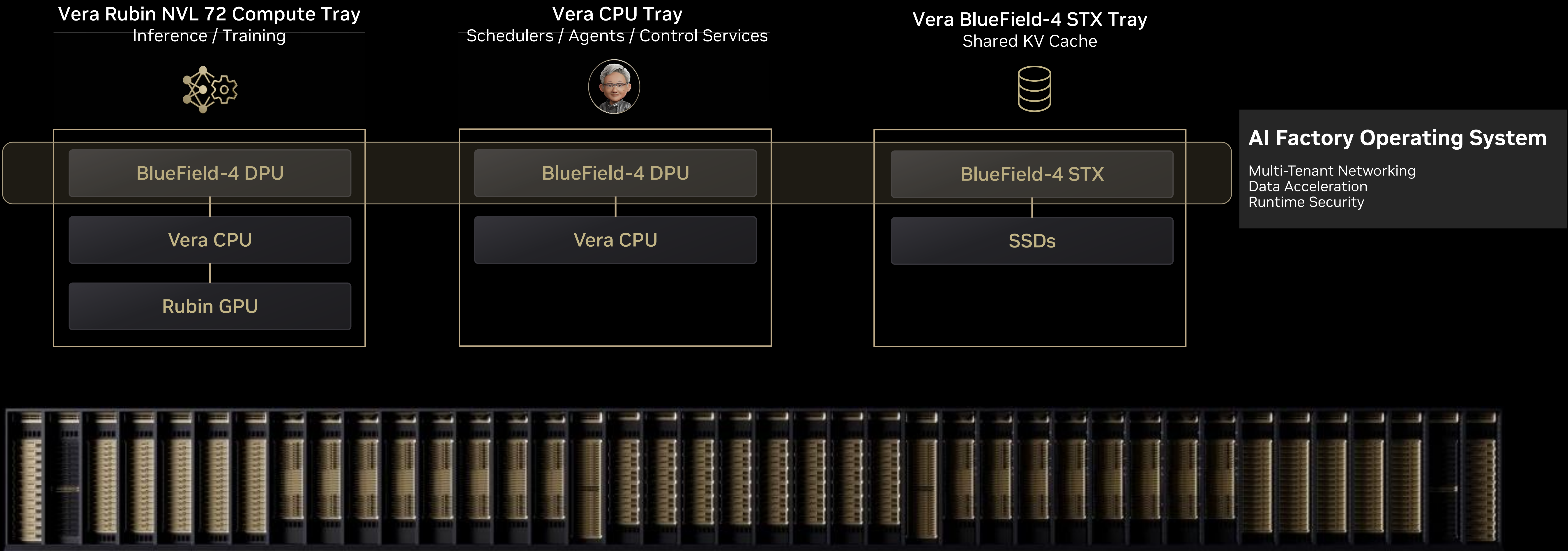
Higher MTBI



NVIDIA Photonics in Production



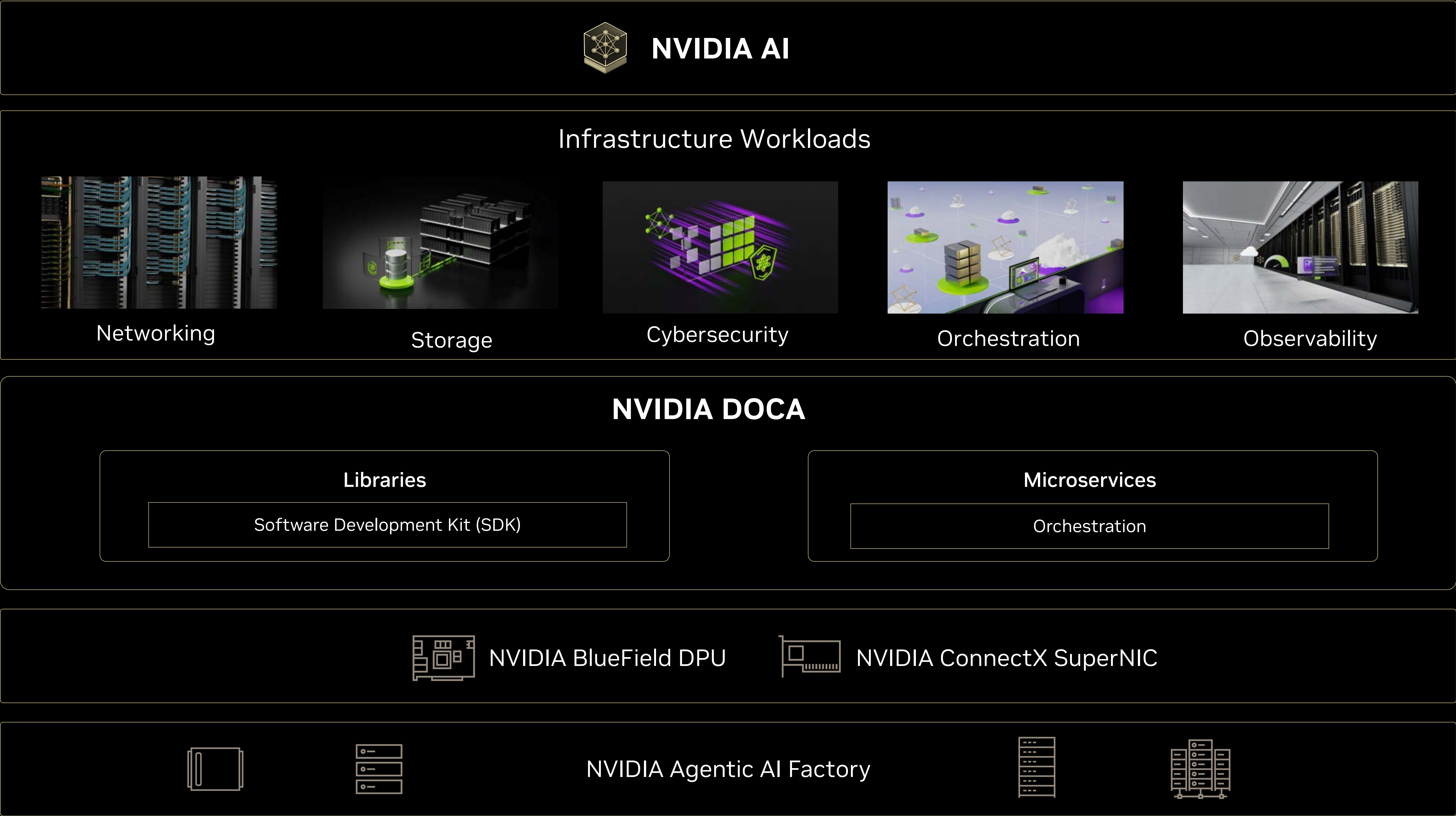
NVIDIA BlueField Powers the Operating System of AI Factories



Vera Rubin AI Factory

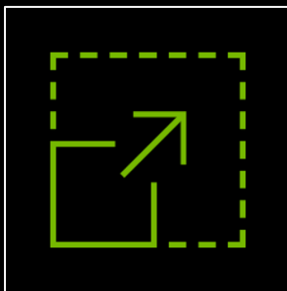
NVIDIA DOCA AI Factories Operating System

Accelerate, secure, and scale agentic AI infrastructure



BlueField-4 CMX: A New Class of AI-Native Storage for Gigascale Inference

Extreme Co-Design for AI-Native Context Memory



Shared across GPUs at SU-scale



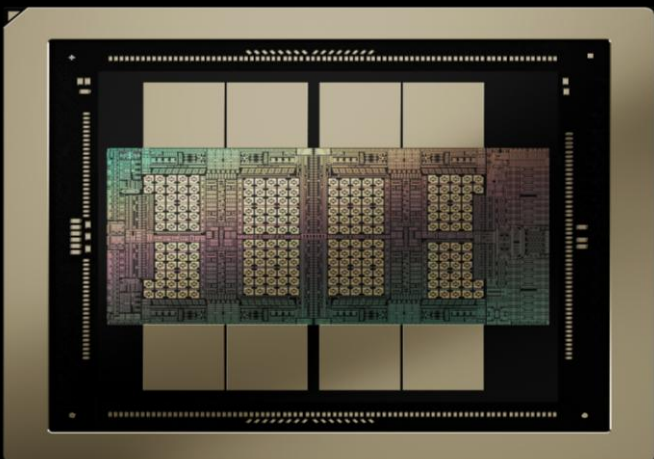
5x higher token throughput



Complements GPU memory & network storage



Co-designed with leading storage partners



Context Memory Hierarchy

GPU HBM	G1 Active KV
System Memory	G2 Staging / Spillover KV
Local Storage	G3 Warm KV cache reuse
Network Storage	G3.5 - CMX
Network Storage	G4 Cold or shared KV context

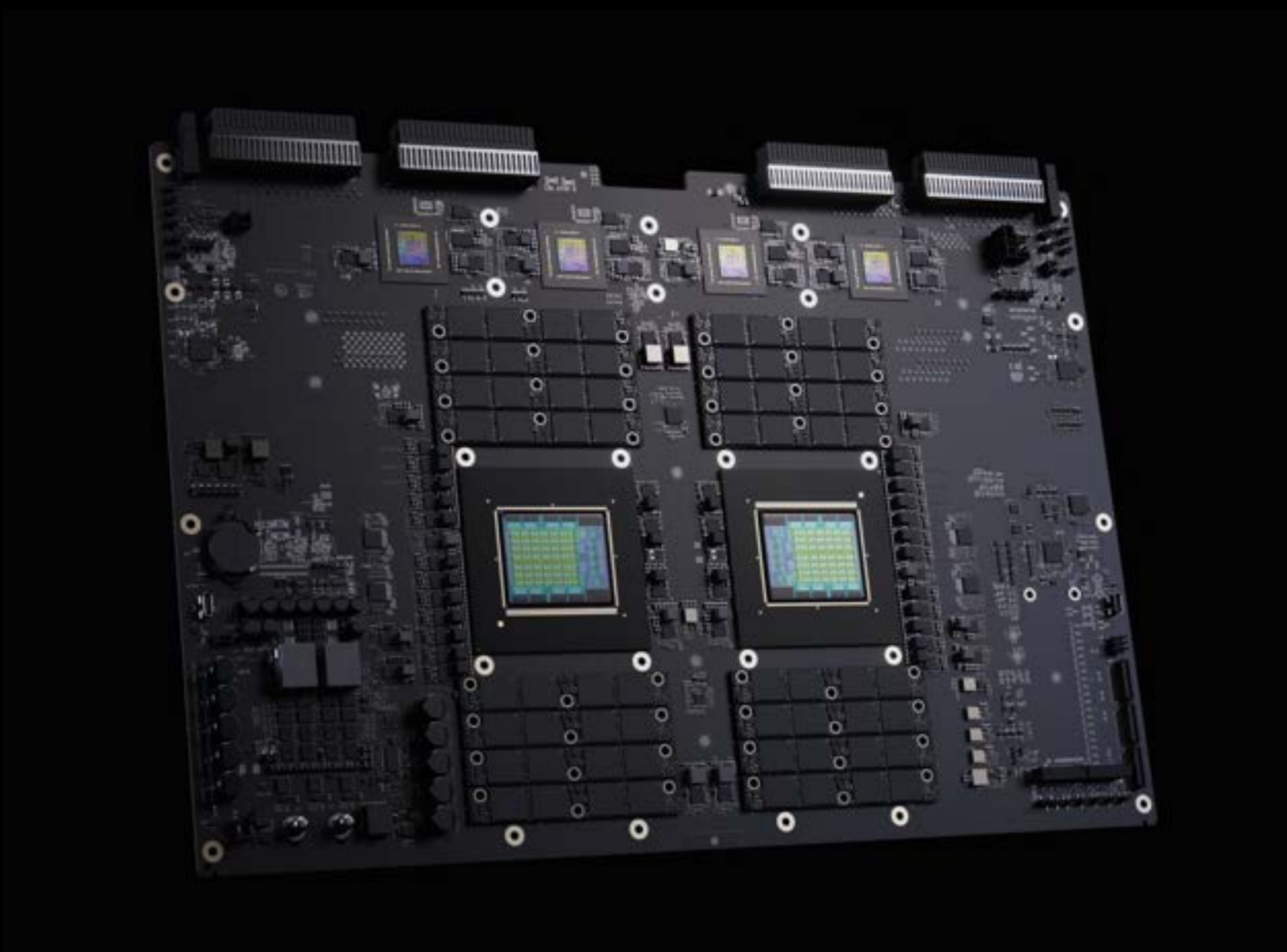
OEM Partners



NVIDIA CMX is Extreme Co-Design for AI-Native Context Memory

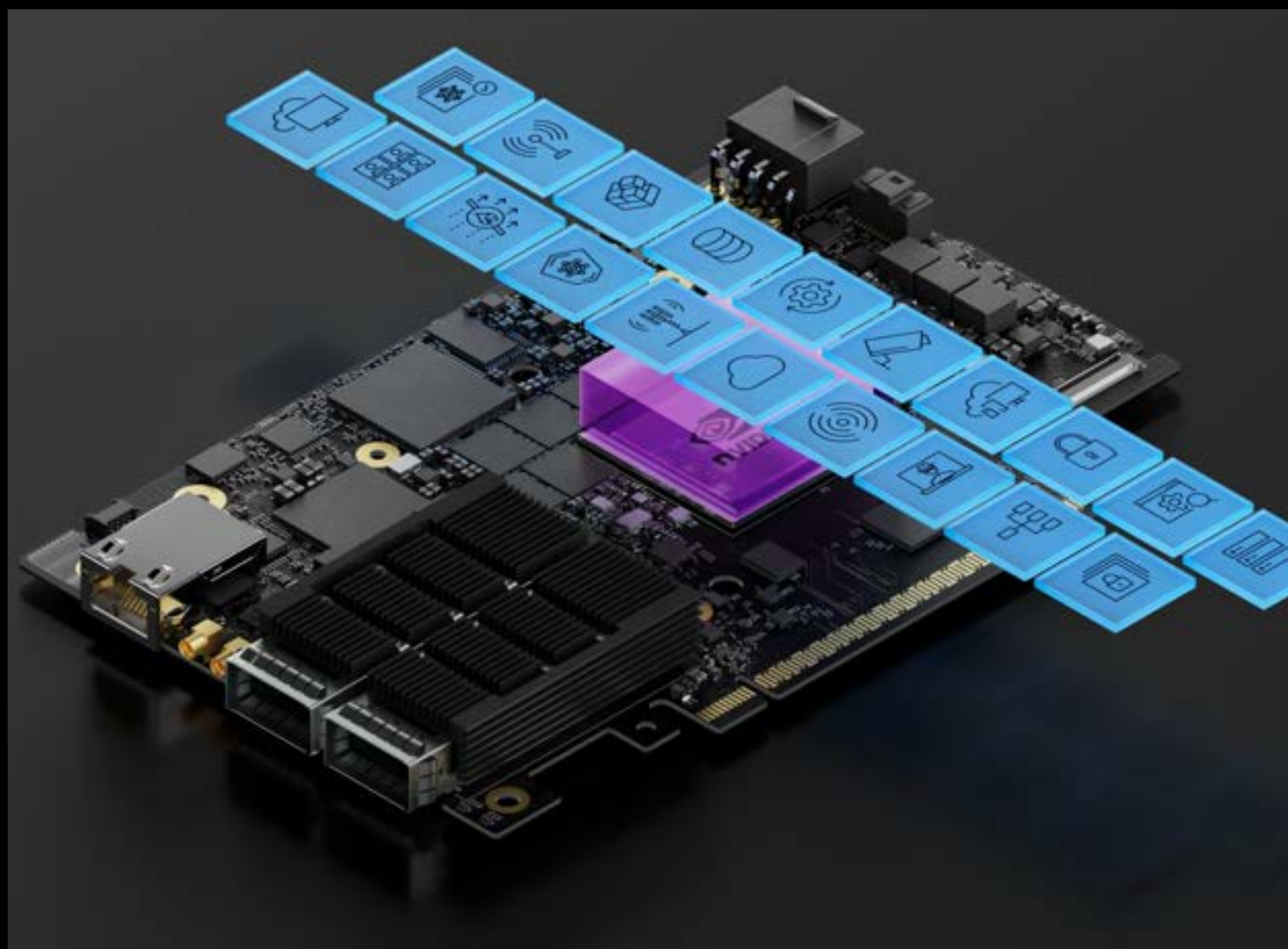
Integrated KV cache processing, acceleration, services, and orchestration

NVIDIA Vera BlueField-4
Storage processor



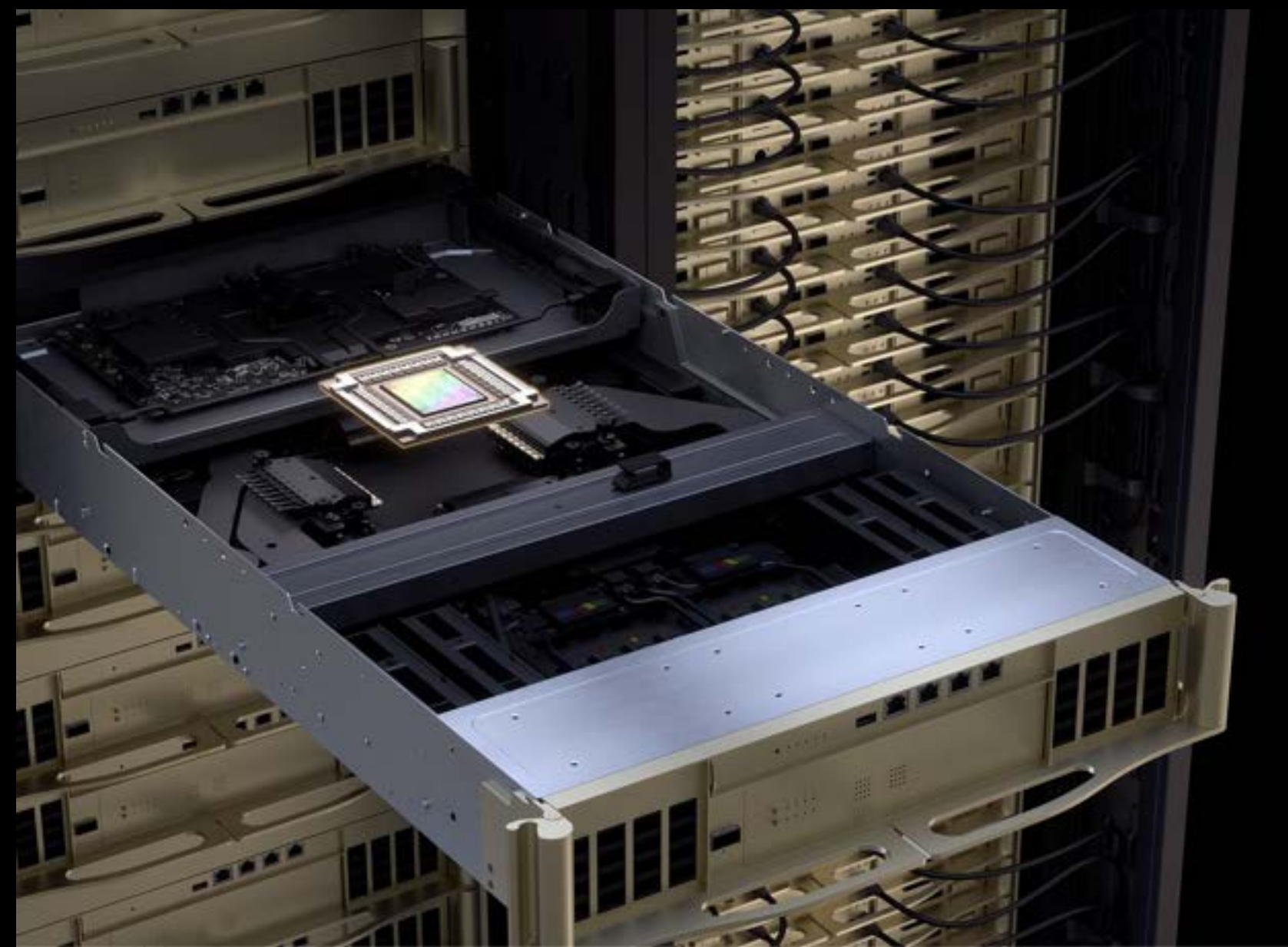
- Runs on AI compute and storage data nodes
- KV cache I/O and storage control
- Turns Ethernet flash into pod-level cache

NVIDIA DOCA
Storage services



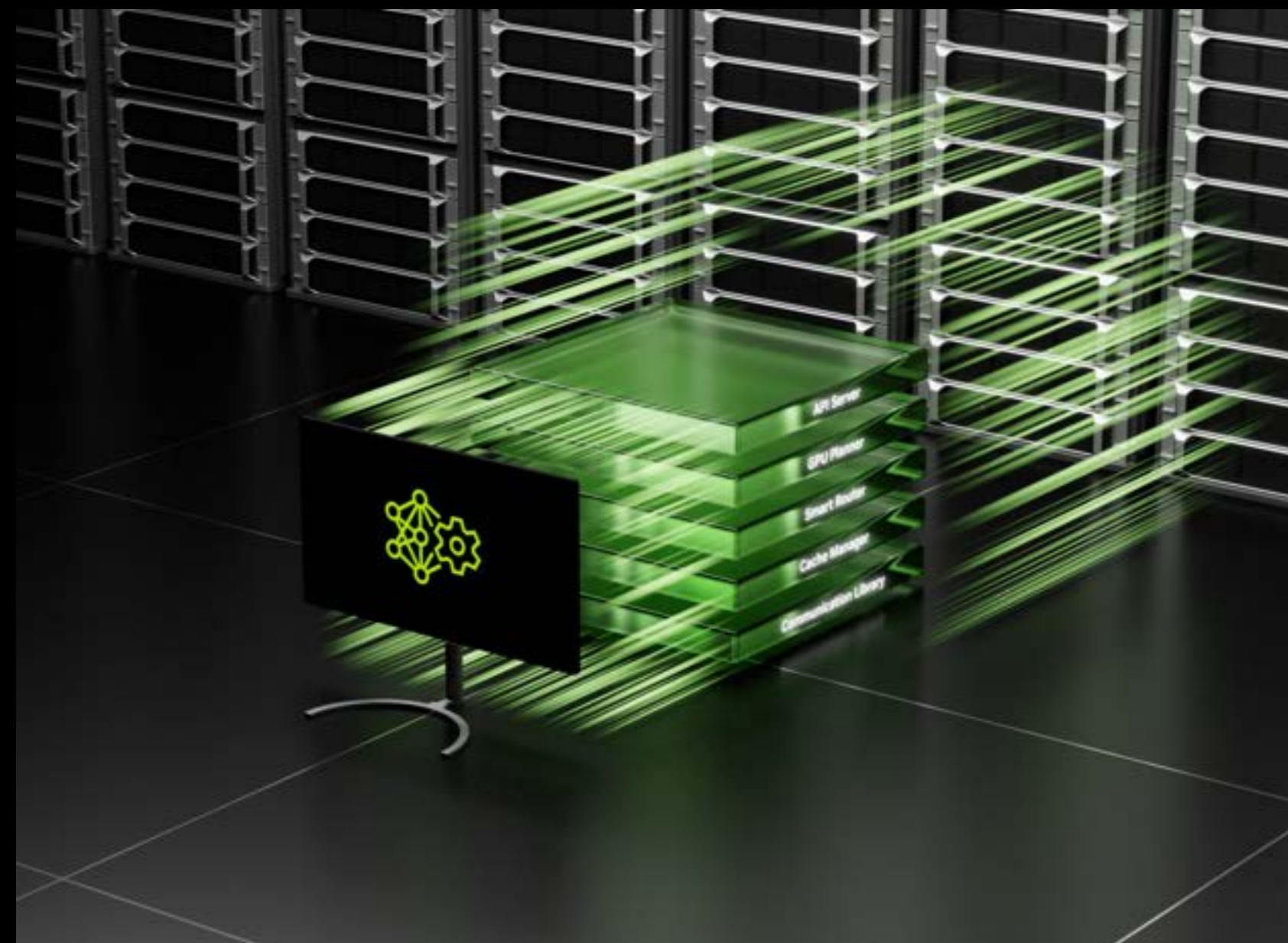
- KV cache-aware services and APIs
- Exposes simple key-value interface
- Data placement, lookup, isolation

NVIDIA Spectrum-X Ethernet
Storage fabric



- Connects GPUs and storage efficiently
- Low-latency RDMA for KV cache sharing
- Predictable bandwidth and low jitter

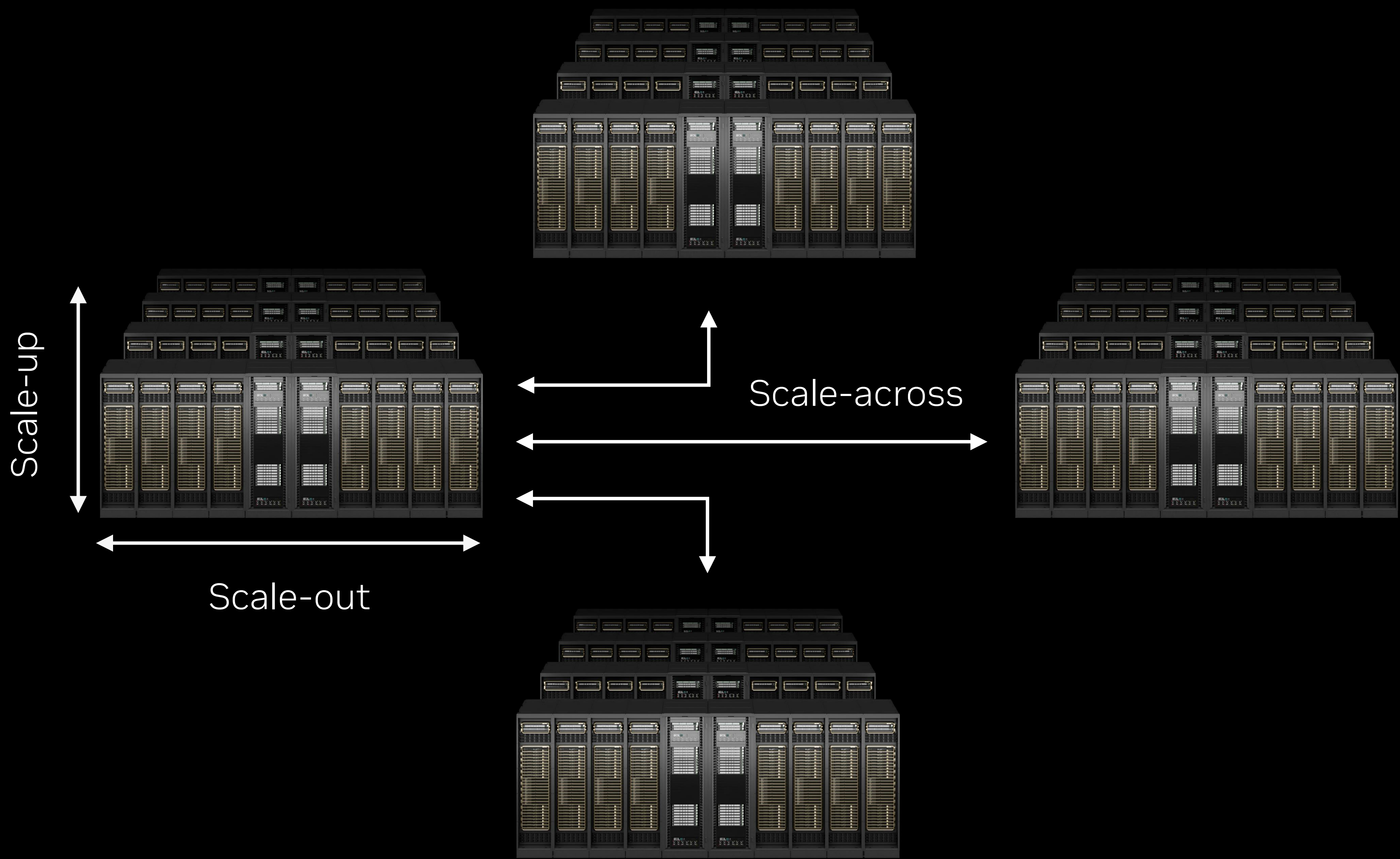
NVIDIA Dynamo
Data orchestration



- Orchestrates inference context across tiers
- Treats context storage as G3.5
- Enables pod-wide context reuse

Spectrum-XGS Ethernet Brings Factories Into Giga-Scale AI

Doubled scale-across performance for multi-site distributed AI operations



1.9x
Multi-site performance

Spectrum-X Ethernet Switch



800Gb/s per port | 102.4Tb/s

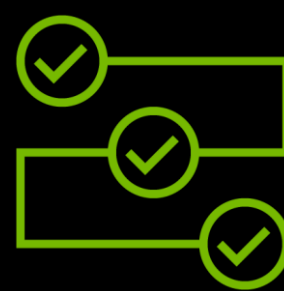
ConnectX-9 SuperNIC



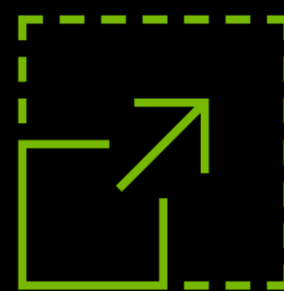
800Gb/s networking



AI Factories
with open
NOS support



Distance-aware
load balancing and
congestion control



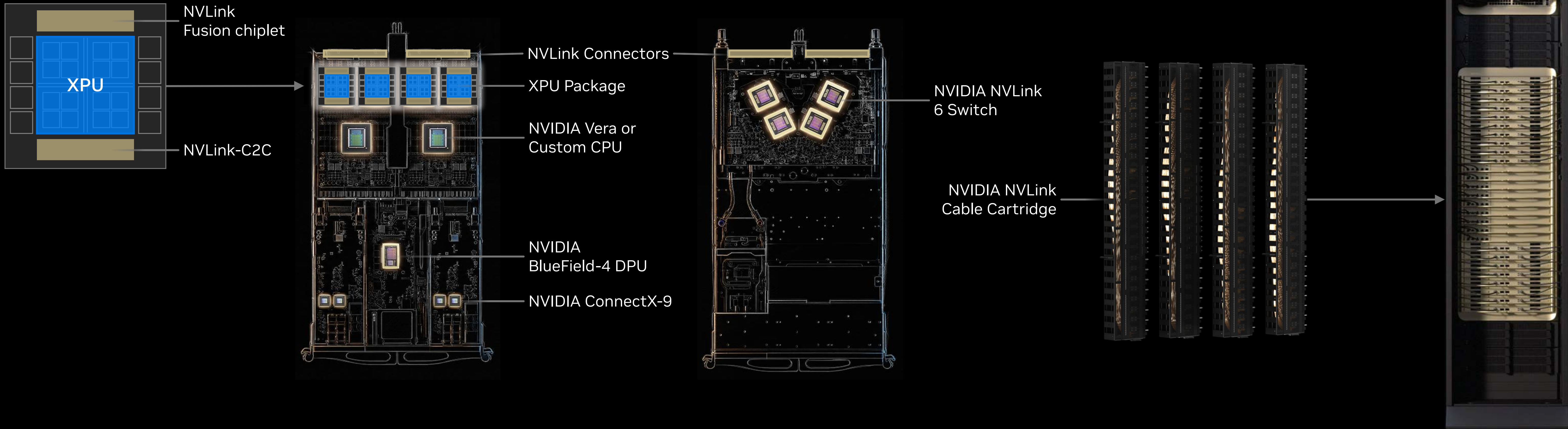
Unlimited scale
across cities,
states, continents



Minimize cross
data-center
latency

NVLink Fusion Connects XPU to the NVIDIA AI Platform

Deploy XPUs with faster TTM, leading NVLink scale-up networking, and unified MGX rack architecture & ecosystem



Compute Tray x18

Cableless, hoseless, modular design
45C coolant inlet temp

NVIDIA NVLink Switch Tray x9

3.6 TB/s per XPU all-to-all bandwidth
3x lower latency
Built-in factory resiliency

NVIDIA NVLink Spine x1

Connects 72 XPUs in a single domain

NVIDIA MGX Rack

Third-generation design
100% liquid cooled

Vera Rubin Platform – Vera Rubin NVL72, Vera CPU Rack, SpectrumX E....



NVIDIA DSX AI Factory Platform

Extreme co-design at infrastructure scale

DSX OS

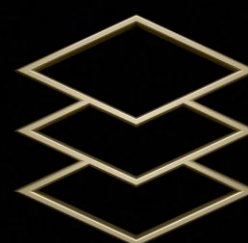
Power Optimization



Infrastructure Software

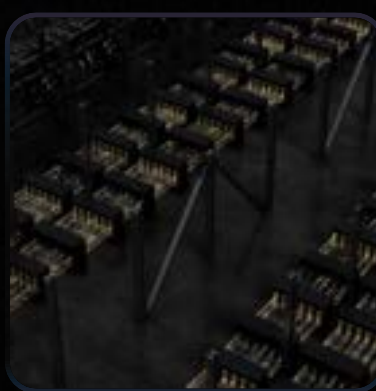


Platform Software



Libraries and Frameworks

DSX Sim



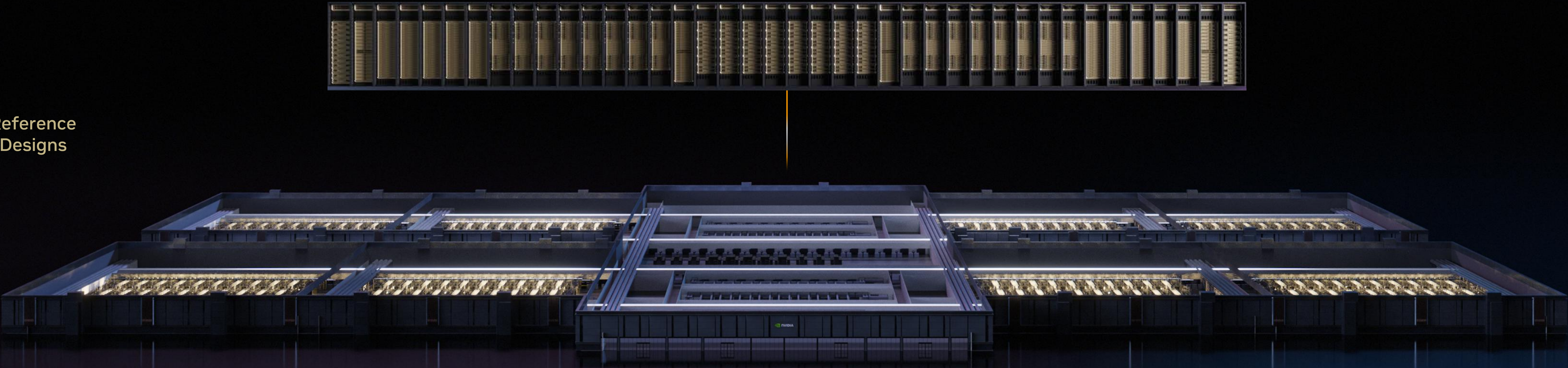
DSX MaxLPS



DSX Flex



Reference Designs



NVIDIA DSX Air

AI Factory design and validation platform

Deploying AI Factories in days instead of weeks

6 Months
to
1 Week

Infrastructure
validation

3 Months
to
1 Week

Software
bring-up

2 Weeks
to
1 Day

Deployment
time



Orchestrator



Storage and Routing



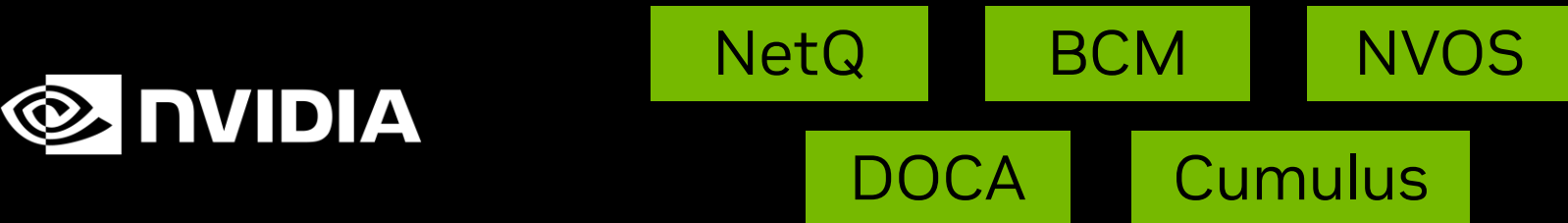
Security



Scheduler



Mission Control



DSX Infrastructure

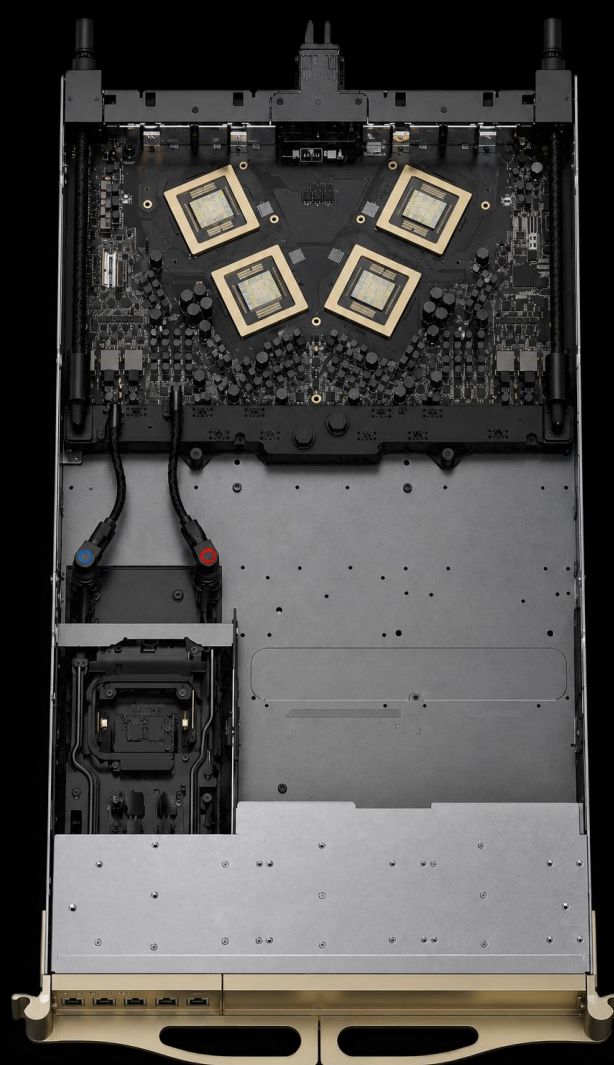


NVIDIA Networking for Giga-scale AI Factories

The only purpose-built networking that scales every dimension for the most efficient and resilient AI factories

Scale-Up

NVLink



10X

Higher
Packet Rate

3X

Lower
Latency

Scale-Out

Spectrum-X Ethernet



1.6X

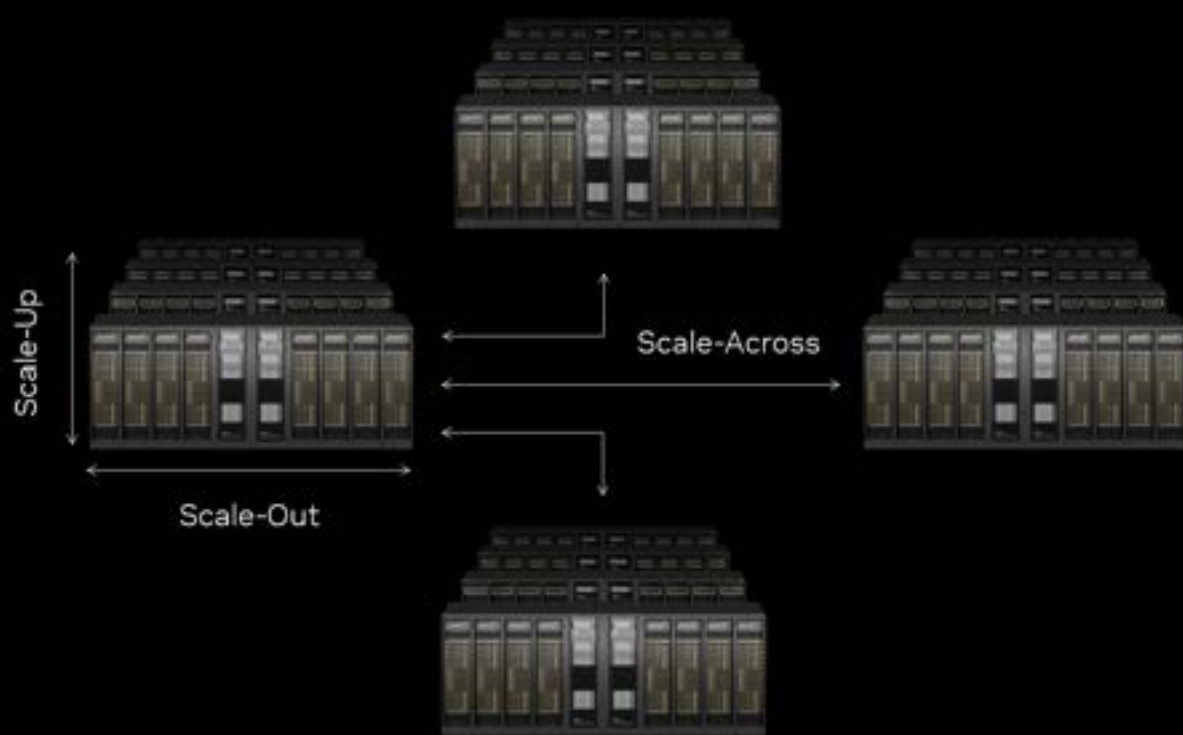
Higher RDMA
Bandwidth

1.3X

Lower
Jitter

Scale-Across

Spectrum-XGS Ethernet



1.9X

Multi-site
Performance

Context Memory

Vera BlueField-4 Storage
Processor



5X

Higher Token
Throughput

5X

Greater Power
Efficiency

